DOXYCAP CAPSULE

VIDOX10-var (MY)

DESCRIPTION
Light green and pink capsule with "DXY 100" printed on one end and "HD" printed on the other.

COMPOSITION
Each capsule contains: Doxycycline hyclate equivalent to Doxycycline 100 mg/ capsule.

PHARMACODYNAMICS
Doxycycline is a semisynthetic tetracycline. It is bacteriostatic and has a wide range of antimicrobial activity. It interferes with bacterial protein synthesis by reversibly binding to bacterial 30S ribosomal subunit, thus preventing the binding of aminoacyl transfer-RNA to the messenger-RNA ribosome complex.

PHARMACOKINETICS
Absorption
Doxycycline is almost completely absorbed and is not subject to presystemic metabolism and the peak serum concentration occurs after 2 to 4 hours. Almost all of the product is absorbed in the upper part of the digestive tract. Absorption is not modified by administration with meals, and milk has little effect.

Distribution
In adults, an oral dose of 200 mg results in;
- A peak serum concentration of more than 3 µg/ml
- A residual concentration of more than 1 µg/ml after 24 hours
- A serum half-life of 16 to 22 hours
- Protein binding varying between 82 and 93% (labile binding) intra- andextracellular diffusion is good.

With usual dosages, effective concentrations are found in the ovaries, uterine tubes, uterus, placenta, testicles, prostate, bladder, kidneys, lung tissue, skin, muscles, lymph glands, sinus secretions, maxillary sinus, nasal polyps, tonsils, liver, hepatic and gallbladder bile, gallbladder, stomach, appendix, intestine, omentum, saliva and gingival fluid. Doxycycline is transferred into breast milk. Only small amounts are diffused into the cerebrospinal fluid.

Biotransformation
No significant metabolism occurs.

Elimination
The antibiotic is concentrated in the bile. About 40 % of the administered dose is eliminated in 3 days in active form in the urine and about 32 % in the faeces.

Urinary concentrations are roughly 10 times higher than plasma concentrations at the same time. In the presence of impaired renal function, urinary elimination decreases, faecal elimination increases, and the half-life remains unchanged. The half-life is not affected by haemodialysis.

INDICATIONS
- Rickettsial infections, including typhus, Rocky Mountain spotted fever and Q fever.
- Chlamydial infections, including psittacosis, lymphogranuloma venereum, trachoma and inclusion conjunctivitis.
- Mycoplasmal infections, especially those caused by Mycoplasma pneumoniae.
- Brucellosis and plague (use in association with streptomycin).
- Tularaemia, chronic bronchitis, chancroid, granuloma inguinale, urinary tract infections, acne.
- Malabsorption syndromes such as tropical sprue and Wipple's disease.
- Cholera, relapsing fever, leptospirosis and early stage of Lyme disease.
- Balantidiasis, amoebic dysentery (use in association with an amoebicide).
- Syphilis, yaws, gonorrhoea, actinomycosis, anthrax, rat bite fever and acute necrotising ulcerative gingivitis.

CONTRAINDICATIONS
- Avoid in patients who have shown hypersensitivity to tetracyclines.

WARNINGS AND PRECAUTIONS
Photosensitivity: Photosensitivity manifested by an exaggerated sunburn reaction has been observed in some individuals taking tetracyclines, including doxycycline. Patients likely to be exposed to direct sunlight or ultraviolet light should be advised that this reaction can occur with tetracycline drugs and treatment should be discontinued at the first evidence of skin erythema.

Use in patients with impaired hepatic function: Doxycycline should be administered with caution to patients with hepatic impairment or those receiving potentially hepatotoxic drugs. Abnormal hepatic function has been reported rarely and has been caused by both the oral and parenteral administration of tetracyclines, including doxycycline.

Use in patients with renal impairment: Excretion of doxycycline by the kidney is about 40%/72 hours in individuals with normal renal function. This percentage excretion may fall to a range as low as 1-5%/72 hours in individuals with severe renal insufficiency (creatinine clearance below 10ml/min). Studies have shown no significant difference in the serum half-life of doxycycline in individuals with normal and severely impaired renal function. Haemodialysis does not alter the serum half-life of doxycycline. The anti-anabolic action of the tetracyclines may cause an increase in blood urea. Studies to date indicate that this anti-anabolic effect does not occur with the use of doxycycline in patients with impaired renal function.

Microbiological overgrowth: The use of antibiotics may occasionally result in over-growth of non-susceptible organisms, including Candida. If a resistant organism appears, the antibiotic should be discontinued and appropriate therapy instituted.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including doxycycline, and has ranged in severity from mild to life-threatening. It is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of antibacterial agents.

Clostridium difficile associated diarrhoea (CDAD) has been reported with use of nearly all antibiotics, including doxycycline, and has ranged in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of C. difficile. C. difficile produces toxins A and B, which contribute to development of CDAD. Hypertoxin producing strains of C. difficile cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD should be considered in all patients who present with diarrhoea after antibiotic treatment. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

Oesophagitis: instances of oesophagitis and oesophageal ulcerations have been reported in patients receiving capsule and tablet forms of drugs in the tetracycline class, including doxycycline. Most of these patients took medications immediately before going to bed or with inadequate amounts of fluid.

Bulging fontanelles in infants and benign intracranial hypertension in juveniles and adults have been reported in individuals receiving full therapeutic drugs. These conditions disappeared rapidly when the drug was discontinued.
Porphyria: There have been rare reports of porphyria in patients receiving tetracyclines.

Venereal disease: When treating venereal diseases, where coexistent syphilis is suspected, proper diagnostic procedures, including dark-field examinations, should be utilised. In all such cases monthly serological tests should be made for at least four months.

Beta-haemolytic streptococci infections: Infections due to Group A beta-haemolytic Streptococci should be treated for at least 10 days.

Myasthenia gravis: Due to a potential for weak neuromuscular blockade, care should be taken in administering tetracyclines to patients with myasthenia gravis.

Systemic lupus erythematous: Tetracyclines can cause exacerbation of systemic lupus erythematosus (SLE).

Jarisch-Herxheimer reaction: Some patients with spirochete infections may experience a Jarisch-Herxheimer reaction shortly after doxycycline treatment is started. Patients should be reassured that this is a usually self-limiting consequence of antibiotic treatment of spirochete infections.

Methoxyflurane: Caution is advised in administering tetracyclines with methoxyflurane.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE
Visual disturbances such as blurring of vision may occur during treatment with doxycycline and in such cases; patients must be informed to refrain from driving or operating machinery.

PREGNANCY AND LACTATION
Use is not recommended during the last half of pregnancy, nursing mothers and children under the age of 8 years as tetracyclines may cause permanent discolouration of the teeth, enamel hypoplasia and inhibition of linear skeletal growth.

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DRUG INTERACTIONS

There have been reports of prolonged prothrombin time in patients taking warfarin and doxycycline.Tetracyclines depress plasma prothrombin activity and reduced doses of concomitant anticoagulants may be necessary.

Since bacteriostatic drugs may interfere with the bactericidal action of penicillin, it is advisable to avoid giving doxycycline in conjunction with penicillin.

Absorption of doxycycline may be impaired by concurrently administered antacids containing aluminium, calcium, magnesium or other drugs containing these cations; oral zinc, iron salts or bismuth preparations. Dosages should be maximally separated.

Barbiturates, carbamazepine, primidone and phenytoin may increase the metabolism of doxycycline (reduced half-life). An increase in the daily dosage of doxycycline should be considered.

Alcohol may decrease the half-life of doxycycline.

The concurrent use of tetracyclines and methoxyflurane has been reported to result in fatal renal toxicity.

Doxycycline may increase the plasma concentration of ciclosporin. Co-administration should only be undertaken with appropriate monitoring.

Drugs that induce hepatic enzymes such as rifampicin may accelerate the decomposition of doxycycline, thereby decreasing its half-life. Sub-therapeutic doxycycline concentrations may result. Monitoring concurrent use is advised and an increase in doxycycline dose may be required.

Ergotamine and methysergide
There is an increased risk of ergotism when doxycycline is co-administered with ergotamine and methysergide.

Methotrexate
Doxycycline increases the risk of methotrexate toxicity; prescribe with caution to patients on methotrexate.

Kaolin and sucralfate may reduce the absorption of doxycycline.

Quinapril contains magnesium carbonate and may interfere with the absorption of doxycycline.

A few cases of pregnancy or breakthrough bleeding have been attributed to the concurrent use of tetracycline antibiotics with oral contraceptives.

There is a possible increased risk of benign intra-cranial hypertension when doxycycline given with retinoids. Concomitant use should be avoided.

Antibacterials inactivate oral typhoid vaccines. Avoid administration of vaccine during treatment with doxycycline.

Laboratory test interactions
False elevations of urinary catecholamine levels may occur due to interference with the fluorescence test.

MAIN SIDE/ADVERSE EFFECTS

The following adverse reactions have been observed in patients receiving tetracyclines, including doxycycline.

Hypersensitivity reactions, including anaphylactic shock, anaphylaxis, anaphylactoid reactions, anaphylactoid purpura, hypotension, pericarditis, angioneurotic oedema, exacerbation of systemic lupus erythematosus, dyspnoea, serum sickness, peripheral oedema, tachycardia and urticaria.

Infections and infestations: As with all antibiotics, overgrowth of non-susceptible organisms may cause candidiasis, glossitis, staphylococcal enterocolitis, pseudomembranous colitis (with Clostridium difficile overgrowth) and inflammatory lesions (with candidal overgrowth) in the anogenital region.

Blood and lymphatic system disorders: Haemolytic anaemia, thrombocytopenia, neutropenia, porphyria and eosinophilia have been reported with tetracyclines.

Immune system diosorders: Jarisch-Herxheimer reaction (frequency not known).

Endocrine disorders: When given over prolonged periods, tetracyclines have been reported to produce brown-black microscopic discoloration of thyroid tissue. No abnormalities of thyroid function are known to occur.

Nervous system disorders: Headache. Bulging fontanelles in infants and benign intracranial hypertension in juveniles and adults have been reported in some individuals receiving full therapeutic dosages of tetracyclines. These are reversible on stopping the drug. Symptoms include blurring of vision, scotomata and diplopia. Permanent visual loss has been reported.

Ear and labyrinth disorders: Tinnitus.

Gastrointestinal disorders: Gastro-intestinal symptoms are usually mild and seldom necessitate discontinuation of treatment. Abdominal pain, stomatitis, anorexia, nausea, vomiting, diarrhoea, dyspepsia and rarely dysphagia. Oesophagitis and oesophageal ulceration have been reported in patients receiving doxycycline. A significant proportion of these cases has occurred with the hydrochloride salt in the capsule form. Tooth discolourationa. Black hairy tongue (frequency not known).

Hepato-biliary disorders: Transient increases in liver function tests, hepatitis, jaundice, hepatic failure and pancreatitis have been reported rarely.

Skin and subcutaneous tissue disorders: Rashes including maculopapular and erythematous rashes occur, exfoliative dermatitis, erythema multiforme, Stevens- Johnson syndrome and toxic epidermal necrolysis, photo-onycholysis. Photosensitivity. Frequency 'rare': Fixed eruption. Drug Rash with Eosinophilia and Systemic Symptoms (DRESS) (frequency not known).

Musculo-skeletal, connective tissue and bone disorders: Arthralgia and myalgia.

Renal and urinary disorders: Increased blood urea.

Reproductive system and breast disorders: Vaginitis.

Reversible and superficial discolouration of permanent teeth has been reported with the use of doxycycline but frequency cannot be estimated from available data

OVERDOSE AND TREATMENT

Clinical features: Nausea, anorexia, diarrhoea, perianal itching, skin eruption, fever, anaphylaxis and liver damage.

Treatment: Emesis or gastric lavage, if appropriate; with the necessary emergency measures, if required.

DOSAGE AND ADMINISTRATION

<u>Adults:</u>
Oral, 200 mg on the first day, then 100 mg once a day.

Acne: Oral, 50 mg daily for 6 - 10 weeks.

Note: Effect of doxycycline on teeth should be considered. Therapy should be continued for at least 10 days in group A beta-haemolytic streptococcal infections.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage:
Store below 30°C. Protect from light and moisture.

Presentation/Packing:
Capsule of 250's & 1000's in bottles, 10 x 10's in blisters.
Not all presentations may be available locally.

Product Registration Holder (Malaysia) /
Product Owner: **HOVID Bhd.**
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